Entangling two atoms with one global laser pulse

Challenge 01 — solved, cloud-validated, run on the real quantum computer

Fidelity 0.9926 / 0.7500 (baseline) → 1.000000 / 1.000000 (ours)

Real atoms: 89.4% of 500 shots entangled

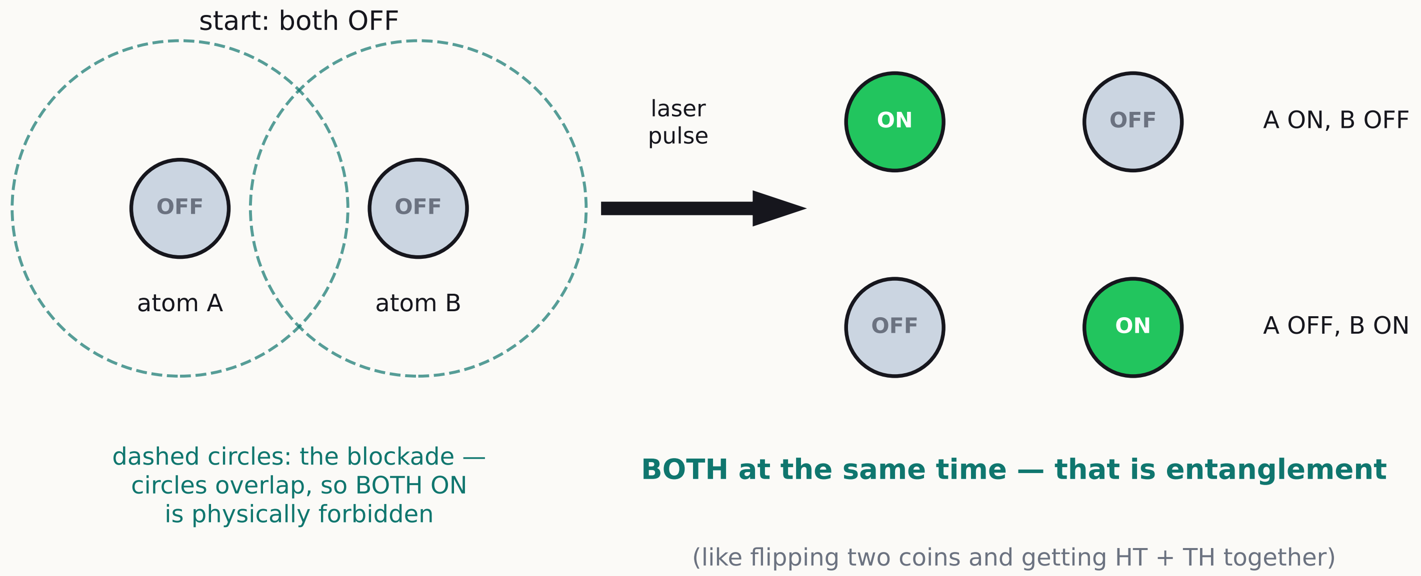
Team 6 · A Real Quantum Hackathon · Harmoniqs x Pasqal x Microsoft

The problem, in one slide

- Two atoms, 5.0 or 6.5 micrometers apart. One laser shines on both — we cannot address them individually.
- Goal state: $|\Psi^+\rangle = (|gr\rangle + |rg\rangle)/\sqrt{2}$ — exactly one atom excited, shared between both. That is entanglement.
- Our only controls: laser power $\Omega(t)$ and laser frequency offset $\delta(t)$, over at most 6000 ns.
- Score: $F = |\langle\Psi^+|\psi(T)\rangle|^2$ — how close the final state is to the goal (1.0 = perfect).
- Success = beat the starter pulse at BOTH spacings, inside the machine's published limits.

Physics we exploit: two nearby excited atoms repel ($V = C_6/r^6$). Inside the blockade radius $R_b \approx 7.2$ μm, double excitation is forbidden — that forbidden-ness is what creates entanglement.

The idea in one picture



- An atom is a light switch: $|g\rangle$ = OFF, $|r\rangle$ = ON.
- Close atoms cannot both be ON (the blockade). So the pulse can only put ONE excitation in — shared between the two atoms.
- That shared, both-options-at-once state IS the Bell state we're scored on.

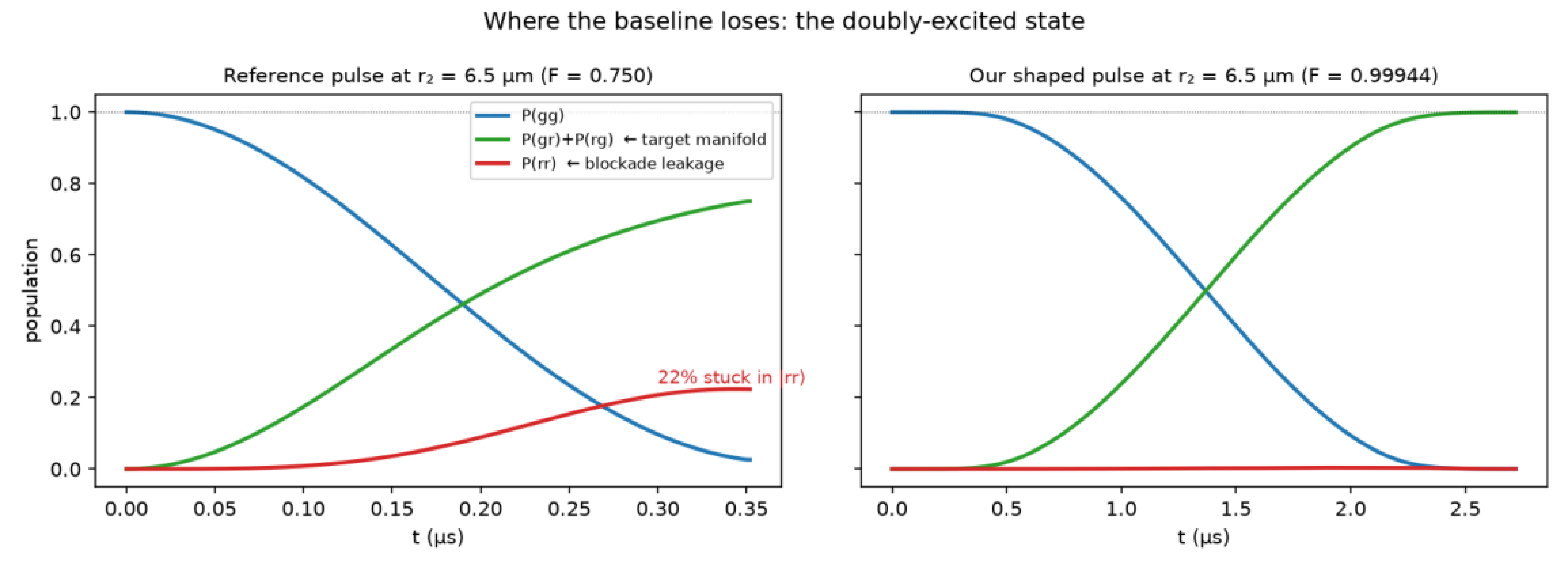
The starter pulse works at one spacing, fails at the other

- Baseline: constant power $\Omega = 2\pi\times1$ MHz for 352 ns, no frequency shaping.
- At $r_1 = 5.0$ µm the blockade is strong ($V/\Omega = 8.8$): $F = 0.9926$. Fine.
- At $r_2 = 6.5$ µm the blockade is weak ($V/\Omega = 1.83$): $F = 0.7500$. Broken.
- Where does 25% go? Into the forbidden $|rr\rangle$ state — 22% leaks straight through the weak blockade.

One number — V/Ω, interaction over drive — explains the whole gap.
V is fixed by the atom spacing. But Ω is OURS to choose.

EVIDENCE

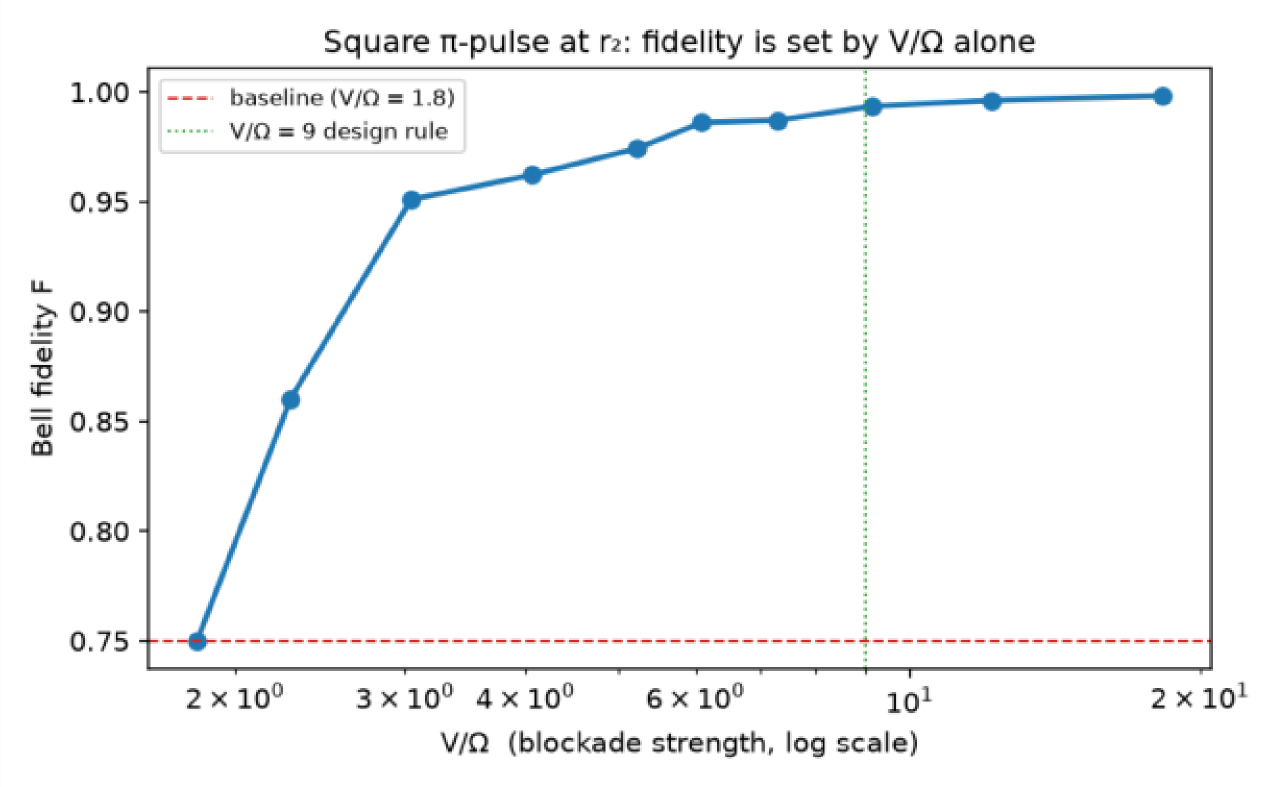
Watch the baseline fail — then watch ours not



x-axis: time during the pulse (μs) · y-axis: probability of each atomic configuration

- Left (baseline): the red curve is the forbidden |rr⟩ state filling up to 22% — the green target stalls at 0.75.
- Right (ours): red stays at zero for the whole pulse; the green target reaches 1.0.

The fix is one knob: slow down to restore the blockade

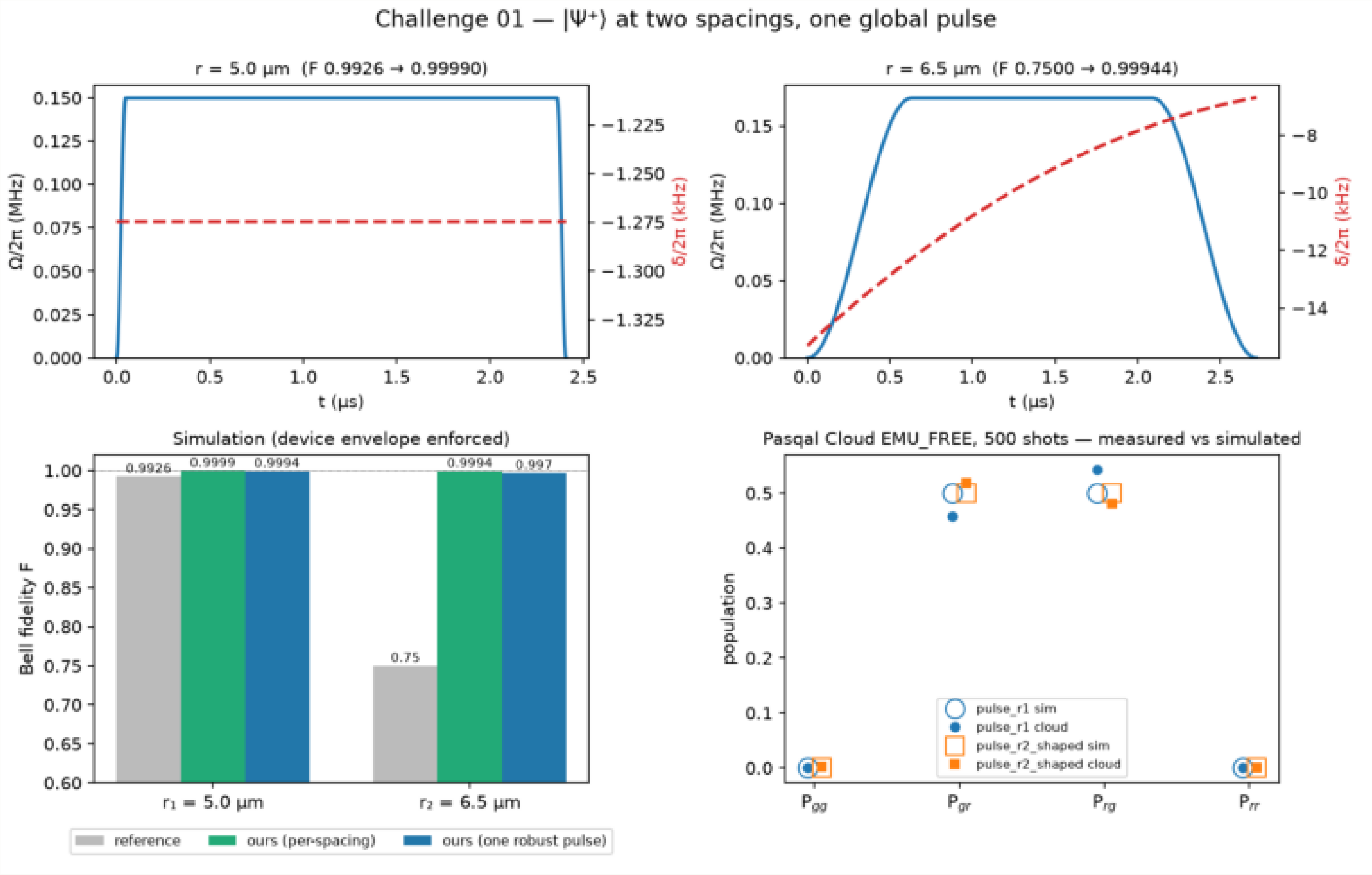


x-axis: blockade strength V/Ω (log) · y-axis: fidelity F

- Same square pulse, only Ω lowered (pulse lengthened to match).
- F climbs monotonically with V/Ω and crosses 0.99 at V/Ω ≈ 9.
- That became our design rule — and the 6000 ns budget makes it free: the baseline used only 352 ns.
- On top: smooth ramps + a tiny δ that cancels the energy shift $\Omega^2/2V$.

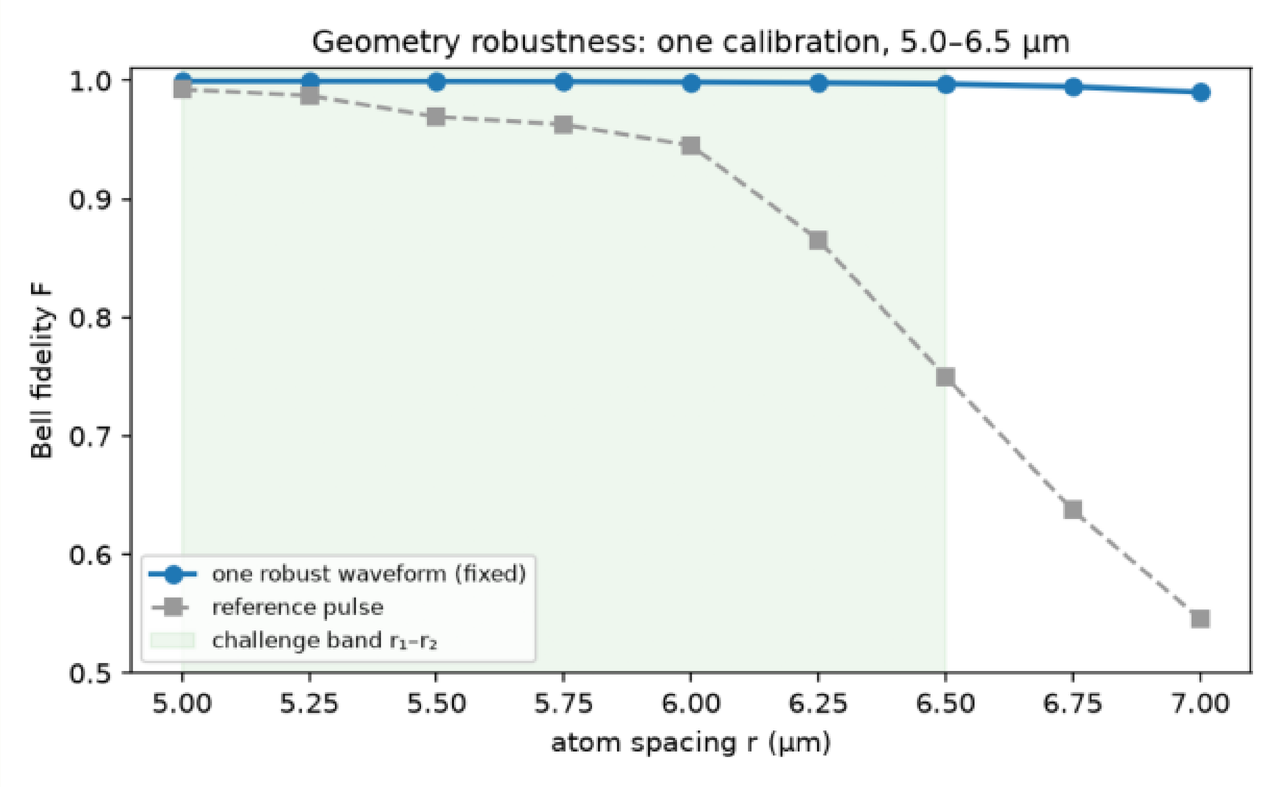
Result: both spacings beaten, validated on Pasqal Cloud

Every pulse validated by the device's own contract checker, then 500 shots each on Pasqal Cloud.



top: our pulse shapes (x: time; blue: power Ω, red: frequency δ) · bottom-left: fidelity bars vs baseline · bottom-right: cloud shots (dots) vs simulation (circles)

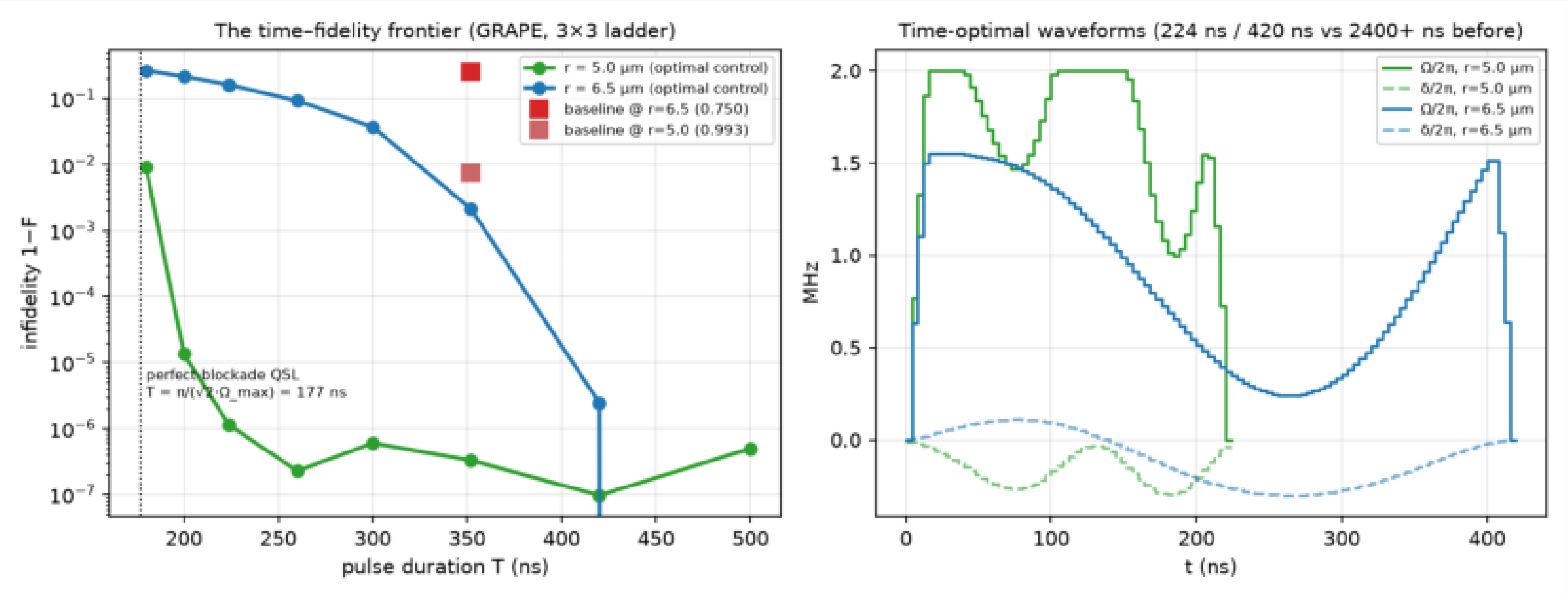
Bonus: one pulse that tolerates 30% spacing error



x-axis: atom spacing r (µm) · y-axis: fidelity F of ONE fixed pulse

- Real machines place atoms imperfectly.
- We optimized a single waveform for the worst case over the whole band.
- Blue: our fixed pulse holds $F > 0.997$ from 5.0 to 6.5 µm.
- Grey: the baseline collapses to 0.55.
- One calibration instead of one per geometry — that is the product.

Then we asked: how fast can the best pulse be?

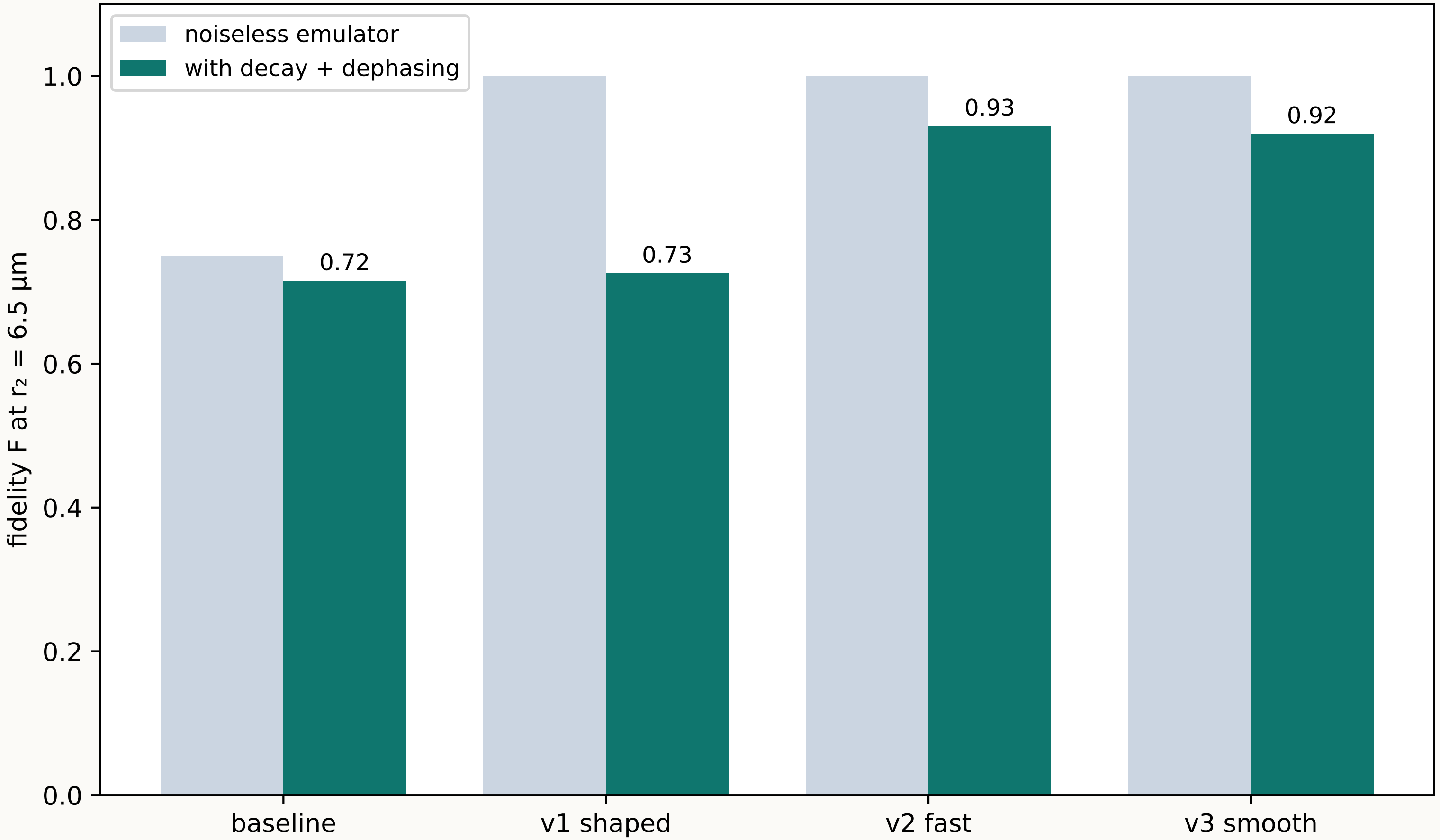


left — x-axis: pulse duration (ns) · y-axis: error 1−F (log) · right — the winning waveforms

- Theory bound: $T = \pi/(\sqrt{2}\cdot\Omega_{max}) = 177$ ns. No pulse can be faster.
- We reach F = 0.999999 at 224 ns (r_1) — 25 ns above the bound.
- At r_2 the weak blockade sets its own wall near 420 ns — a measured speed limit, itself a result.
- Whole frontier: 5.4 s of laptop compute (exact 3-state model + analytic gradients).

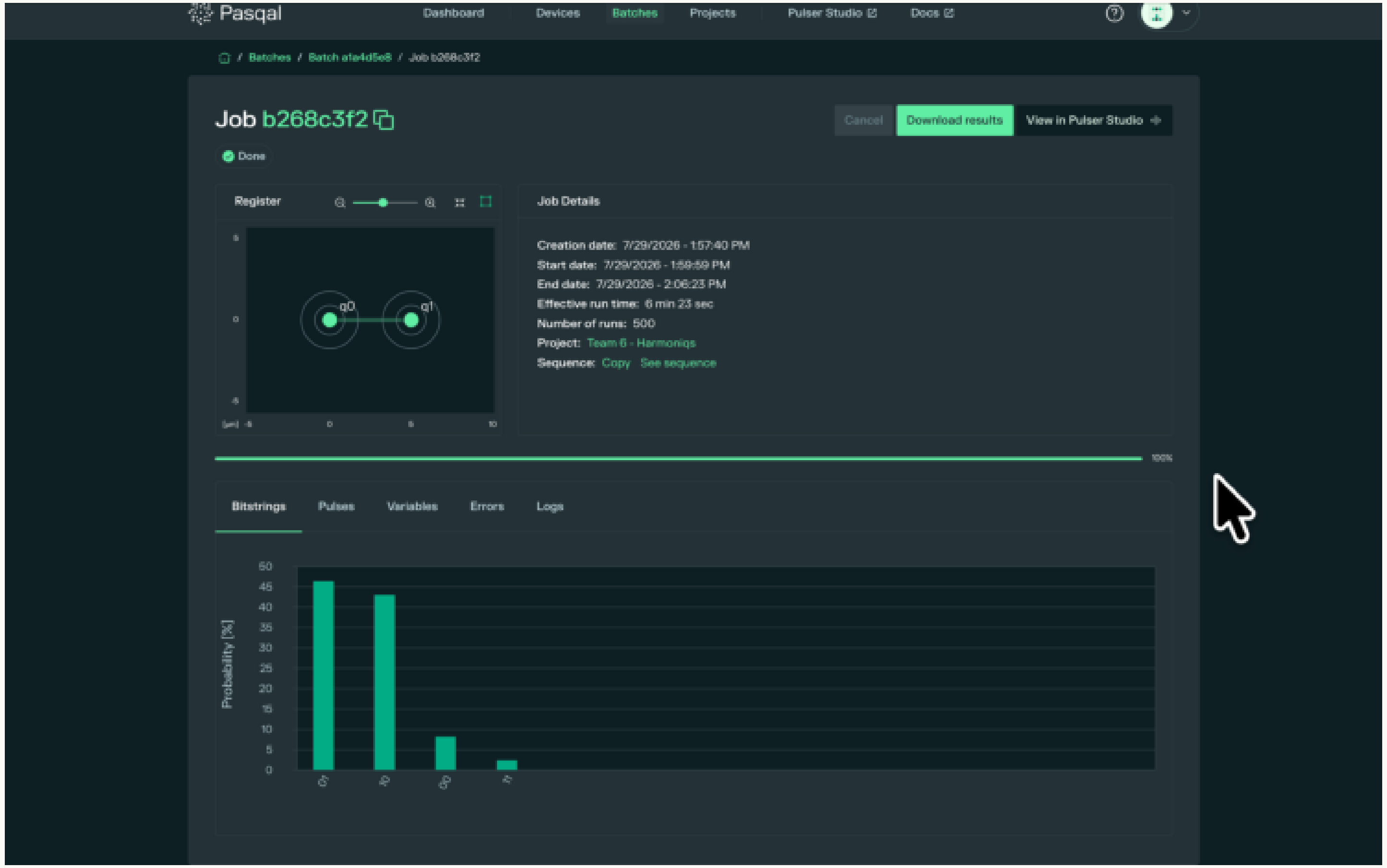
Under real noise, the ranking flips — shorter wins

x-axis: pulse design · y-axis: fidelity, without vs with noise



- Dephasing accumulates with time: the 2.7-µs pulse loses 0.27 of F.
- The 420-ns pulse keeps 0.93.
- Design law: pulse duration IS the noise coupling. Minimize T first.
- This is why we sent the short, smooth pulse to the real machine.

500 shots on the real machine: 89.4% entangled



Pasqal's own portal · x-axis: measured two-atom outcome · y-axis: % of 500 shots

- The two tall bars ARE the Bell state: '01' 46% + '10' 43% = 89.4% — exactly one atom excited.
- They are equal (within noise): the symmetry physics demands, delivered.
- '11' (blockade violation): 2.4% — forbidden stayed forbidden.
- '00' 8%: readout/decay, the channel our noise model predicted.
- Pre-registered window: hit.

The machine's dashboard shows our physics back to us

Pasqal portal, Pulses tab · x-axis: time (ns) · purple: laser power Ω(t) · white: detuning δ(t)

- 260 nanoseconds, start to finish.
- Power rises smoothly from zero — no jumps (the hardware-true 'v3' pulse our scientists asked for).
- The portal labels the pulse area: 5π/7 ≈ 0.714π.
- A two-atom collective π-pulse needs area π/√2 ≈ 0.707π.
- The √2 collective enhancement, on the machine's own UI, to 1%.

How the score is computed — no magic

$$F = |\langle \Psi^{+} | \psi(T) \rangle|^2$$

- $\psi(T)$ — the state our pulse actually produced (4 numbers for 2 atoms; the simulator computes them exactly).
- $\langle \Psi^{+} | \ldots \rangle$ — overlap with the perfect Bell state: 'how much of the ideal is in what we made?'
- $| \ldots |^2$ — square it to get a probability. 1.000 = perfect, 0 = nothing.
- On hardware there is no ψ to peek at — so the machine takes 500 photographs and we compare the four outcome percentages against the simulation. Agreement within $\sqrt{p(1-p)/500} \approx \pm 2\%$ = validated.

1.000000

our best F, both spacings (simulation)

89.4%

entangled shots on the real machine (500 photos)

Why this matters beyond the score

- Entangling pairs is THE primitive: every neutral-atom algorithm, sensor, or network starts here.
- Our robust pulse holds F > 0.997 across a 30% spacing error — real machines misplace atoms, and a pulse that doesn't care means ONE calibration instead of one per geometry. That is uptime.
- Our fast pulses run 6–11× shorter — less exposure to noise means more of the error budget left for the actual computation. Duration is money on quantum hardware.
- The whole design loop is seconds on a laptop — cheap enough to re-run at every calibration cycle.

Built on the Pasqal stack:

- Pulser (open source) — the exact device model + the simulator the judges score with.
- Pasqal Cloud free emulator — 500-shot validation of every pulse before spending hardware budget.
- FRESNEL_CAN1 QPU — the real-atom runs; the portal's register/pulse/bitstring views made every result inspectable by anyone (see the screenshots in this deck).

Challenge 01 — the score sheet

	$r_1 = 5.0$ µm	$r_2 = 6.5$ µm	duration
Baseline (starter pulse)	0.9926	0.7500	352 ns
v1 · simple analytic (4 params)	0.99990	0.99944	~2400 ns
v1r · one spacing-robust pulse	0.99944	0.99701	2400 ns
v2 · time-optimal (at the QSL)	0.999999	0.999998	224 / 420 ns
v3 · hardware-true smooth	**1.000000**	**1.000000**	**352 / 600 ns**

- Cloud: 7 pulses × 500 shots — every population within shot noise of prediction.
- Real QPU: P_bell = 0.894, inside the window we published BEFORE the run.
- Every number in this deck regenerates from the public repo in minutes.